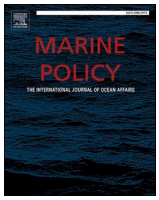
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Benchmarking sustainable blue economy in Canada and Norway: A comparative KPI-based assessment

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ABSTRACT

Since 2015, Canada and Norway have advanced sustainable blue economy (SBE) agendas through distinct policy pathways. This study provides a comparative assessment of their SBE trajectories using a harmonized key performance indicator (KPI)-based framework across four operational dimensions: blue economic growth, circular economy adoption, nature-based solutions (NBS), and carbon alignment. Drawing on a 20-indicator comparative grid aligned with SDGs, OECD guidance, and the EU *Restore Our Ocean and Waters* agenda, the study assesses progress between 2015 and 2025 using an indicative five-point scale. The results show that Norway performs more strongly in blue economic growth and maritime decarbonization, reflecting its mature ocean industries, integrated governance, and early green-shipping transition. Canada performs more strongly in NBS and shows substantial progress in circular economy governance, driven by rapid gains in marine protection, restoration, and plastics policy. The comparison reveals complementary strengths rather than a single model of SBE implementation: Canada offers lessons in collaborative conservation and restoration, while Norway provides a reference case for maritime industrial transition and low-emission shipping. The findings highlight the importance of institutional context, policy sequencing, and implementation capacity in shaping SBE outcomes, and identify transferable lessons for marine policy design, comparative benchmarking, and sustainable ocean governance beyond the two cases.

1. Introduction

Since Rio+ 20 in 2012, the blue economy has gained prominence as a framework for linking ocean-based development with sustainability objectives. Yet the concept is not used in a single or neutral way. The literature shows that the blue economy is interpreted through competing discourses that variously emphasize natural capital, economic growth, conservation, livelihoods, justice, and the legitimacy of marine sectors. In this sense, blue economy narratives are both influential and contested, especially when they are mobilized to justify different governance priorities and development pathways [1,2]. At the European level, this debate has been further shaped by a policy shift formalized by the European Commission in May 2021, which explicitly called for a transition from a "blue growth" paradigm to a "sustainable blue economy" (SBE), aligning maritime sectors with the objectives of the European Green Deal.

With growing international attention to ocean governance, these debates are gaining importance. Since 2015, frameworks such as the Sustainable Development Goals (especially SDG 14), the Paris Agreement, and the EU's *Restore Our Ocean and Waters* Mission have reinforced calls for more integrated approaches to marine conservation, pollution reduction, climate alignment, and sustainable ocean development [1,3]. These frameworks emphasize marine ecosystem conservation, pollution reduction, and climate alignment [4]. In parallel, organizations such as the OECD (*Organization for Economic Co-operation and Development*) and the Ocean Panel (*High-Level Panel for a Sustainable Ocean Economy)* have promoted policy approaches that combine ocean-based growth with stronger governance, natural capital

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protection, and low-carbon transitions [5,6]. Recent work also highlights that sustainable ocean development extends beyond economic and environmental performance to encompass broader social, institutional, and scientific dimensions [7]. At the same time, ocean governance scholars have warned that blue economy transitions can reproduce exclusionary decision-making and uneven distributions of benefits and burdens if justice, inclusion, and participation are not adequately addressed [8].

Within this broader agenda, Canada and Norway provide a particularly relevant comparison. Both are major ocean nations with long coastlines, strong maritime interests, and active participation in international ocean governance. Both joined the Ocean Panel in 2018 and committed to putting sustainability at the core of ocean management [9]. However, they have followed distinct policy trajectories. Canada, with its extensive maritime territory, has more recently accelerated its SBE agenda through initiatives such as the *Oceans Protection Plan* (OPP), the development of a whole-of-government *Blue Economy Strategy* (BES), expanded marine conservation targets, and coastal restoration funding [10–14]. Norway, by contrast, has a longer-standing maritime policy tradition and is widely recognized as a "leading ocean nation," with established strengths in fisheries, aquaculture, shipping, offshore industries, and, increasingly, maritime decarbonization [15–20]. Comparing these two countries is therefore analytically useful not only because both are influential ocean economies, but also because they represent contrasting governance histories, implementation styles, and policy sequencing within the broader SBE agenda.

This comparison is also timely from a policy perspective. International frameworks such as the EU's *Restore Our Ocean and Waters by 2030* mission and OECD guidance on sustainable ocean economies increasingly call for action across several fronts, including economic development, circularity, ecosystem restoration, and decarbonization [2,6,21]. Canada and Norway have both advanced policies in these areas, but their relative progress remains uneven across sectors and dimensions. Despite the growing number of national blue economy strategies, policy reports, and thematic studies, few works benchmark Canada and Norway side by side through a harmonized comparative framework [22–24]. Existing analyses often remain nationally focused, sector-specific, or conceptually broad without offering an operational basis for cross-country assessment.

This paper addresses that gap by comparing Canada's and Norway's progress across four operational dimensions of the SBE: blue economic growth, circular economy adoption, nature-based solutions (NBS), and carbon alignment. These dimensions do not claim to exhaust the full normative scope of the SBE. Rather, they are used here as a policy-relevant and documentable subset that recurs across contemporary ocean policy frameworks and can support structured cross-country benchmarking. This narrower operationalization is particularly useful for comparative policy analysis, while acknowledging that social equity, institutional quality, and scientific capacity remain essential complementary dimensions for future work [7,8].

Methodologically, the paper uses a harmonized key performance indicator (KPI)-based approach to compare how each country translates SBE ambitions into policy commitments, implementation measures, and observable outcomes. KPI-based systems are useful for structuring complex assessments, but the broader performance-measurement literature also warns that indicator-based approaches can oversimplify qualitative realities and may not fully capture hard-to-measure dimensions [25,26]. For this reason, the present study treats the KPI framework as an interpretive but systematic benchmarking tool, rather than as a claim to exhaustive or purely objective measurement.

Accordingly, this paper asks: How do Canada and Norway compare in their progress across four core dimensions of the SBE, and what policy lessons emerge from a harmonized KPI-based assessment? To answer this question, the study pursues three objectives: (i) to develop an operational comparative framework for benchmarking selected SBE dimensions; (ii) to assess Canada's and Norway's performance across the four selected dimensions; and (iii) to derive policy-relevant lessons for strengthening sustainable ocean governance in both countries and beyond. By doing so, the paper contributes to the growing literature on SBE assessment while offering a structured benchmarking approach that may be adapted to other national contexts.

2. Methodology

2.1. Analytical framework and KPI grid

This study adopts a comparative case-study design combining policy analysis with indicator-based performance assessment. Comparative case studies are appropriate when the objective is not only to describe national trajectories, but also to identify similarities, differences, and transferable policy lessons across institutional settings. Canada and Norway were selected because they are both advanced ocean nations with significant maritime sectors, strong international visibility in ocean governance, and active engagement in sustainable ocean policy processes, while also representing distinct development pathways and governance traditions within the broader SBE agenda. This combination makes them suitable for structured comparison.

The analytical framework is based on a harmonized KPI grid covering four operational dimensions: blue economic growth, circular economy adoption, NBS, and carbon alignment in coastal and marine sectors (Fig. 1). These dimensions were selected because they recur across major contemporary ocean policy frameworks and capture core areas through which SBE ambitions are translated into policy and practice. At the same time, they should be understood as an operational subset, rather than a complete representation of the full normative scope of the SBE. Broader sustainable ocean development frameworks also emphasize social, institutional, and scientific dimensions, which were not fully operationalized here but remain important for future research [7,8].

Under blue economic growth, the selected KPIs include blue economy GDP contribution, employment, and sectoral diversification, including activity in emerging sectors such as marine renewable energy and marine biotechnology [27]. Under circular economy adoption, the KPIs cover marine litter reduction, recycling and recovery programs in fisheries and shipping, end-of-life management, and innovation related to waste-to-product pathways. The NBS dimension includes marine protected area (MPA) coverage, habitat restoration, and the integration of ecosystem-based and nature-based approaches into climate adaptation and coastal management. The carbon alignment dimension includes maritime greenhouse gas emissions, clean maritime transport deployment, offshore renewable energy development, and climate-smart infrastructure and policy commitments.

The KPI grid was designed to balance policy relevance, cross-country comparability, and data availability. This reflects a common trade-off in performance-measurement systems: indicators must be sufficiently structured to support comparison, while also acknowledging that complex governance processes cannot be fully reduced to purely quantitative metrics [25,26]. In this study, the KPI framework is therefore used as a structured benchmarking tool for comparative policy analysis, not as a claim to exhaustive or purely objective measurement.

2.2. Scoring approach

Each KPI was evaluated using a five-point ordinal scale (0–5) representing the degree of policy development and implementation observed between 2015 and 2025, where 0 indicates no clear evidence of policy action or progress and 5 indicates a leading or highly advanced level of progress relative to current international expectations. Half-point increments were used where evidence fell between two adjacent scoring levels. Scores were assigned based on triangulated evidence from policy documents, government reports, project data, and secondary literature.

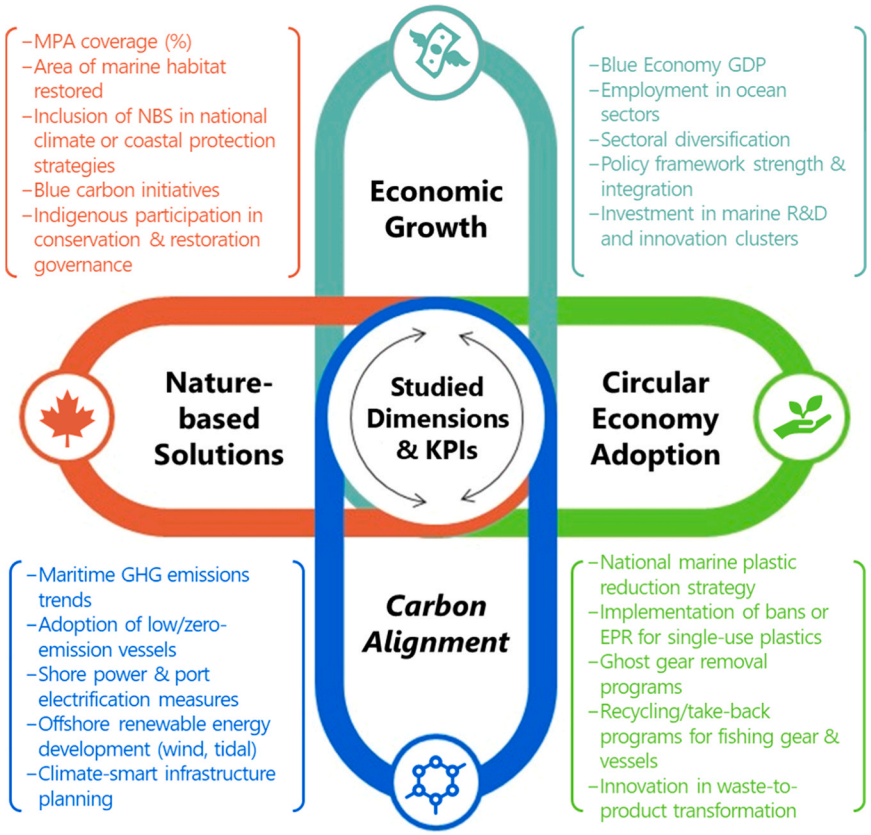


Fig. 1. KPI Grid for SBE assessment. Source: Authors' synthesis based on the international SBE frameworks and literature discussed in Section 2.1.

The scoring process considered three types of evidence: (i) policy intent and formal commitments; (ii) implementation measures and governance instruments; and (iii) observable outputs or outcomes where data were available. For example, under the NBS dimension, an MPA-related KPI received a higher score where a country demonstrated not only formal area-based protection targets, but also measurable expansion, implementation mechanisms, and evidence of ecological or governance integration. A score of 5 for MPA coverage corresponded to very advanced progress, such as approaching or exceeding internationally recognized targets with evidence of enforcement or strong implementation, whereas a score of 1 indicated only limited or early-stage action. Similarly, under carbon alignment, higher scores were assigned where policy commitments were accompanied by tangible deployment of low- or zero-emission maritime technologies, regulatory instruments, or sector-wide transition strategies.

Individual KPI scores were then aggregated at the dimension level to provide a comparative overview and support radar-chart visualization. These aggregated scores should be interpreted as indicative comparative assessments, rather than precise numerical rankings. Their function is to synthesize mixed qualitative and quantitative evidence in a transparent and comparable way. KPI definitions and scoring thresholds are presented in Table S1, while the detailed comparative scores are reported in Table S2 (Appendix).

2.3. Data collection

The assessment drew on a combination of peer-reviewed literature, official government documents, policy strategies, project-level reports, and selected grey literature. For Canada, core sources included the BES engagement paper, the What We Heard report, OPP annual reports, departmental reports, and official press releases on major milestones such as the launch of BES consultations and the release of the *Blue Economy Regulatory Roadmap* (BE2R). For Norway, the analysis used English versions of major policy documents and white papers, including *New Growth*, *Proud History* (2017), *Blue Opportunities* (2019), and thematic materials related to green shipping, marine litter, and ocean-climate policy. International sources, including OECD ocean economy reports and European Commission blue economy reports, were used to contextualize national trends and improve cross-country interpretation.

To strengthen the empirical basis of the comparison, the analysis also incorporated project-level and program-level evidence. For example, the *Canadian Coastal Restoration Fund* (CRF) was examined as a concrete indicator of restoration ambition and implementation, while Norwegian initiatives such as kelp forest restoration pilots were considered as evidence of emerging restoration practice. Likewise, circularity analysis included national plastic strategies, fisheries-related recovery programs, and marine waste initiatives in both countries. Quantitative information, such as GDP contribution, employment estimates, MPA coverage, restoration areas, or electric vessel deployment, was cross-checked against multiple sources wherever possible.

2.4. Data analysis and synthesis

The analytical procedure combined document review, evidence synthesis, and comparative scoring. First, relevant evidence from each country was collected and organized by the selected dimensions. Second, each KPI was assessed using the proposed scoring approach, based on the combined consideration of policy commitments, implementation instruments, and observable outputs. Third, country-level evidence was synthesized into comparative narratives to explain not only relative scores, but also the policy pathways underlying them.

The analysis used 2015 as a common baseline and 2025 as the reference endpoint for assessing change. The year 2015 was selected because it aligns with the consolidation of the contemporary SBE agenda through SDG 14 and related international sustainability frameworks, while 2025 provides a relevant near-term policy horizon linked to

current national and international commitments. Using this time window made it possible to assess progress over a decade characterized by intensified ocean policy activity in both countries. Where direct quantitative equivalence between countries was not possible, the interpretation relied on reasoned comparison of policy trajectories and available outputs. For instance, differences in national definitions of ocean economy, differences in conservation instruments, and differences in governance structures were considered during interpretation. Qualitative elements such as stakeholder engagement, governance integration, and collaborative approaches were incorporated where relevant to explain observed policy performance, while remaining secondary to the main operational KPI framework.

2.5. Validation and limitations

This study has several limitations. Although the KPI framework provides a structured basis for comparison, it simplifies a complex and multidimensional policy field and may underrepresent qualitative, relational, and context-specific aspects of performance, especially where outcomes are difficult to quantify directly [25,26]. The scoring also remains partly interpretive. To reduce arbitrariness, scores were based on triangulated evidence from multiple sources, explicit scoring logic, and internal consistency checks, but they should still be understood as indicative rather than definitive measurements. Inter-coder reliability was not formally tested, which remains a methodological limitation. Data comparability between Canada and Norway is also imperfect, as the two countries do not always define or report ocean economy indicators in the same way and some policy domains are supported by richer datasets than others. As a result, some KPIs rely more heavily on policy and implementation evidence than on strictly comparable quantitative metrics. In addition, social equity and justice were not included as a standalone dimension. This reflects a deliberate narrowing of scope toward dimensions that were more consistently documentable across the two cases, rather than a judgment that equity is of secondary importance. Future studies should build on this framework by integrating social, institutional, and distributional dimensions more explicitly [7,8]. Finally, the findings are most directly applicable to advanced ocean economies with strong institutional capacity and active engagement in international ocean governance; they should therefore be seen as analytically transferable rather than statistically representative.

3. Results

3.1. Overall comparative performance

Fig. 2 presents the aggregate scores for the four assessed dimensions. Norway scores higher in blue economic growth, circularity, and carbon alignment, reflecting its larger and more mature ocean economy and its more advanced maritime decarbonization trajectory. Canada performs more strongly in NBS and remains competitive in circularity, driven by rapid progress in marine conservation, restoration funding, and plastic-pollution policy. Overall, both countries display relatively balanced profiles, but with different areas of comparative advantage. Detailed KPI evidence is summarized in Table S2 (Appendix).

3.2. Blue economic growth

Norway's blue economy remains a major pillar of national economic performance. In 2019, ocean industries generated NOK ≈ 694 billion in value creation and employed ≈ 225,000 people, representing ≈ 30% of private-sector value creation and 11% of employment. Since 2015, Norway has reinforced its strategic commitment to ocean-based growth through the 2017 Ocean Strategy and the 2019 Blue Opportunities update, which emphasize aquaculture, offshore wind, marine minerals, digitalization, and ocean technology [28,29]. Growth in salmon farming has further strengthened Norway's global position in seafood exports [30]. Although sectors such as tourism and petroleum were affected during the COVID-19 period [31,32], recovery efforts were closely

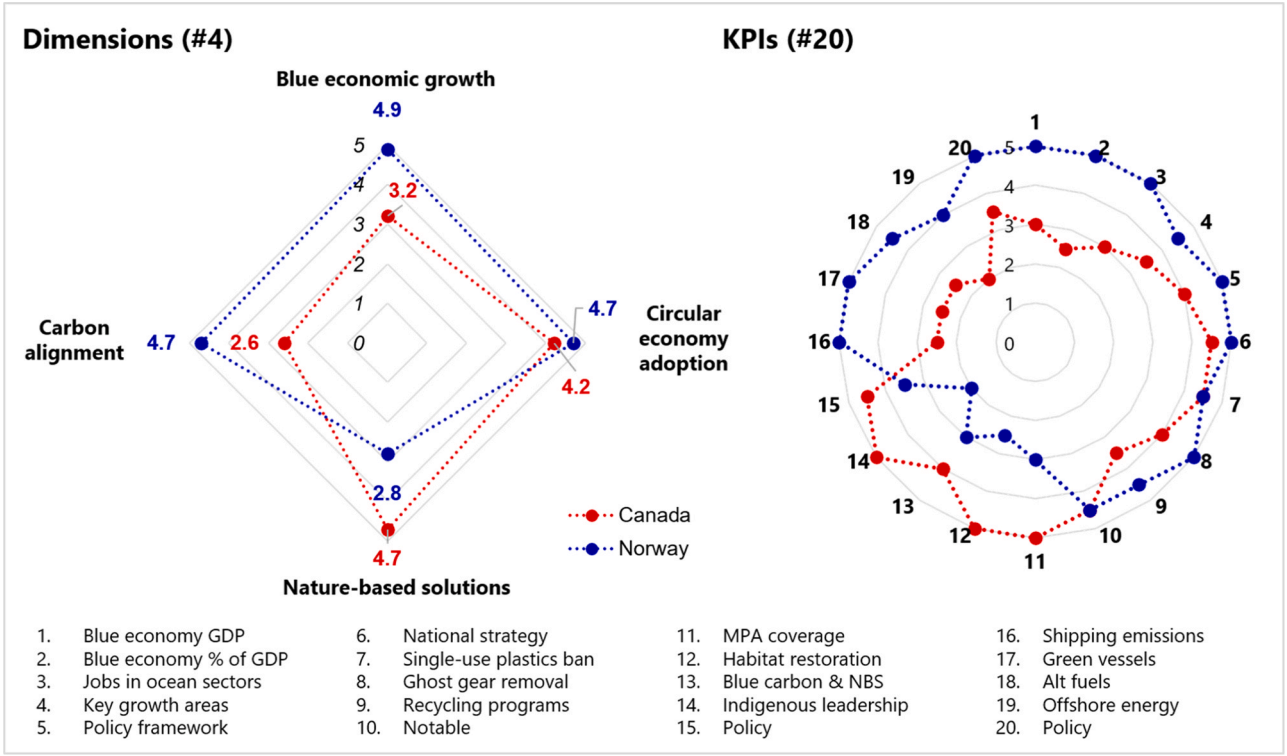


Fig. 2. Radar chart showing Canada's and Norway's SBE implementation (2015–2025). Higher scores indicate better performance. Norway leads in blue economic growth and carbon alignment, while Canada leads in NBS, both countries perform comparatively strongly in circularity. Source: Authors' calculations based on the KPI scores reported in Table S2 (Appendix).

linked to green-transition investments, including low-emission shipping. Norway's long-term planning, public-private coordination, and integrated marine spatial planning framework continue to support its leadership position in this dimension [20,33].

Canada's blue economy is smaller in relative terms, accounting for ≈ 1.6% of GDP [34], although it remains economically significant, generating CAD ≈ 31.7 billion annually and supporting around 300,000 jobs [11]. Since 2015, Canada has increasingly relied on policy-led growth instruments, including investments in marine safety, oil-spill prevention, coastal restoration, and innovation [35]. A notable example is the Ocean Supercluster, launched in 2018, which has co-funded more than 60 innovation projects valued at over CAD 360 million [36]. Canada also retains strong traditional sectors, particularly seafood, while exploring emerging activities such as offshore wind and tidal energy. However, expansion remains more constrained in some areas, notably aquaculture [37]. Recent policy efforts, including the BES and the 2024 Regulatory Roadmap, aim to reduce barriers in marine renewables, spatial planning, and ocean technology while linking growth to Indigenous and coastal-community participation. Overall, Norway outperforms Canada in this dimension because of the greater scale, maturity, and diversification of its ocean economy, whereas Canada's trajectory remains more recent and policy driven.

3.3. Circular economy adoption

Norway has positioned itself as an early and relatively strong performer in integrating circularity principles into marine policy [38]. Its 2017 plastics and circularity strategy helped structure subsequent measures targeting marine litter, recycling, and port-based waste management. Norway also played an active international role in marine plastics diplomacy, including support for global plastics action and contributions to international funding mechanisms. Domestically, its measures include high bottle-recycling rates, mandatory port waste sorting, zero-discharge rules in parts of the offshore sector, and incentives for fishing and boating waste return schemes [39]. Although some gaps remain, especially regarding ship recycling outside national territory, Norway shows relatively mature implementation across several circularity-related indicators.

Canada has also made substantial progress, especially since 2018 [40]. During its G7 presidency, it launched the Ocean Plastics Charter and subsequently advanced the Zero Plastic Waste Strategy and related action plans. These initiatives set ambitious targets for plastic recovery and recycling and reflect a clear shift toward circular-economy framing in marine pollution policy. Canada also committed CAD 100 million to support plastic waste management in developing countries, reinforcing its international profile. However, implementation remains uneven due to the country's federal structure and the strong role of provincial and municipal authorities in waste governance. While early evidence suggests some reduction in single-use plastics on Canadian coastlines, the long-term effects of bans and circularity policies remain emergent.

Both countries perform strongly in this dimension, but Norway retains a slight advantage because it moved earlier and has more mature operational mechanisms in place. Canada, however, has become highly visible in global plastics governance and has developed a robust policy architecture that may yield stronger implementation results over time. As shown in Table S2 (Appendix), both countries have advanced national plastic reduction strategies and ghost-gear actions, although extended producer responsibility remains more developed in some areas than others.

3.4. Nature-based solutions

Canada performs strongly in the NBS dimension, particularly in marine conservation and restoration. Since 2015, its MPA coverage has expanded from about 1–13.8%, with targets of 25% by 2025 and 30% by 2030 [41]. This expansion includes new MPAs and strengthened minimum protection standards restricting activities such as oil and gas, bottom trawling, mining, and waste dumping. Canada has also invested heavily in habitat restoration. Between 2017 and 2022, the CRF supported 64 projects, restoring about 650 km^2 of aquatic habitat and creating about 1300 jobs [42,43]. Additional programs, including the Aquatic Ecosystems Restoration Fund and the Natural Infrastructure Fund, further reinforce this trajectory. Canada's NBS profile is also strengthened by Indigenous-led and co-managed conservation initiatives in Arctic, Pacific, and Great Lakes contexts [44].

Norway's NBS profile is more mixed. It has a long tradition of ecosystem-based marine management and precautionary planning, but its formal MPA coverage has lagged Canada's. By 2021, Norway had designated 17 MPAs and expanded other protected coastal areas, while continuing to emphasize broader ecosystem-based management and area-based fisheries measures as key conservation tools [45]. Restoration is growing but remains less mature and less visible institutionally than in Canada. Current efforts include kelp forest recovery, blue forest research, salmon habitat restoration, and localized coastal wetland use for flood mitigation.

A growing sub-pillar of NBS involves the valorization of maintenance-dredged sediments [46–48]. In Canada, projects such as Maplewood Marine Restoration and the Shippagan Sand Engine pilot show how dredged materials can support habitat creation and coastal resilience. In Norway, large-scale remediation efforts such as those in the Oslofjord have demonstrated the feasibility of clean capping but have not yet been clearly linked to habitat creation.

3.5. Carbon alignment in coastal and marine sectors

Norway performs more strongly on carbon alignment, largely because it combines ambitious climate policy with concrete deployment in maritime decarbonization [49]. Through procurement requirements, incentives, and sectoral innovation policies, Norway has accelerated the uptake of electric and hybrid ferries and other low-emission vessels, while also supporting broader transition efforts in domestic shipping and related infrastructure [50]. Norway has also backed international measures, including carbon pricing discussions within the IMO (*International Maritime Organization*), and supported low-emission maritime transitions abroad [51]. These actions make Norway a widely cited reference case for coastal and maritime decarbonization [30].

Canada's progress is more gradual. Although the country has committed to economy-wide emissions reduction and net-zero targets, implementation in the marine sector remains less advanced than in Norway. Important developments are underway, including shore-power expansion, planned ferry electrification, and participation in green shipping corridor initiatives, but deployment remains at an earlier stage and more fragmented across jurisdictions and policy instruments [52].

Overall, Norway's stronger performance in this dimension reflects the closer alignment between policy ambition, industrial strategy, and technological deployment [53]. Table 1 summarizes the main carbon-alignment pathways in both countries across key coastal and marine sectors.

3.6. Cross-dimensional comparative synthesis

Table 2 synthesizes these comparative strengths and the main analytical insights emerging across the four assessed dimensions. The cross-dimensional comparison reveals two distinct SBE implementation pathways. Norway's performance is anchored in mature maritime industries, coordinated governance, and deployment-oriented policy instruments, whereas Canada's progress is driven more strongly by recent public programs, conservation initiatives, and collaborative governance. Neither pathway produces uniform leadership across all dimensions, highlighting the importance of institutional context and policy sequencing.

Table 1
Comparative carbon-alignment pathways across key coastal and marine sectors.

Theme	Norway	Canada
Low-emission vessels and fuels	First battery-electric ferry (*Ampere*, 2015); strong green procurement for ferry decarbonization; large electric/hybrid ferry deployment; shore power at major ports.	Hybrid and LNG-powered ferries in operation; early battery-electric ferry deployment; decarbonization efforts supported by the OPP and broader marine-fleet strategies; shore power expanding at major ports.
Regulation and incentives	Zero-emission requirements for cruise ships and ferries in UNESCO-listed fjords from 2026; 2020 *Greener and Smarter Shipping* white paper; national procurement and policy support for low-emission maritime transport.	Emission Control Area standards for nitrogen and sulfur oxides reduction; incentives through Green Marine and public funding; marine decarbonization is increasingly embedded in climate plans and low-carbon shipping initiatives.
Offshore energy transition	Floating offshore wind development; electrification of offshore oil and gas platforms using renewable-powered subsea connections; strong R&D in wave, ocean, and hydrogen-linked marine energy.	Offshore wind emerging, especially in Atlantic Canada; offshore oil and gas decarbonization remains limited; early hydrogen and port-energy initiatives under development.
Ports and infrastructure	Shore-side electricity available at major ports; The Longship carbon capture and storage project links maritime logistics with subsea CO_2 storage; port and energy infrastructure support low- to zero-emission shipping.	Shore power and electrified port infrastructure expanding; federal fleet modernization includes hybrid vessels and biofuel use; National Trade Corridors Fund supports lower-carbon port development.
Blue carbon and climate-related measures	Blue carbon and coastal climate measures are increasingly linked to resilience and mitigation; carbon capture and storage and marine climate infrastructure play a visible role in the broader low-carbon transition.	Blue carbon is increasingly integrated into reporting, research, and coastal policy; wetlands, salt marshes, and related NBS are emphasized in coastal climate adaptation.

4. Discussion

The comparison indicates that differences between Canada and Norway are best explained by policy sequencing, institutional configuration, and sectoral path dependence rather than by differences in stated ambition alone. Norway's performance reflects the integration of sustainability objectives into a mature maritime-industrial system, whereas Canada's trajectory is shaped more strongly by recent public programs, conservation expansion, and collaborative governance. These contrasting pathways illustrate how national political economies influence the operationalization of SBE.

4.1. Drivers of progress and comparative advantages

The comparison highlights three main drivers of divergence across the assessed SBE dimensions. First, policy sequencing appears to be a major explanatory factor. Norway entered the current SBE period with a mature ocean economy, established maritime clusters, and an existing governance architecture capable of linking industrial development with marine planning. This institutional foundation enabled sustainability objectives to be integrated into established industrial policies, planning systems, and implementation mechanisms. Canada, by contrast, has advanced through more recent strategic programs and public investments, including the OPP, restoration funds, and the BES process, within a comparatively fragmented policy landscape. Second, institutional configuration shapes implementation capacity. Norway's more integrated marine management framework appears to reduce fragmentation and support stronger coordination between economic and environmental objectives [54]. Canada's more federalized governance structure can slow policy harmonization, but it also creates space for regional experimentation, Indigenous-led initiatives, and multi-actor collaboration.

Table 2
Comparative synthesis of Canada and Norway across the four assessed SBE dimensions.

Dimension	Canada	Norway	Comparative insight
Blue economic growth	Recent policy-led expansion supported by public investment, innovation programs, and a whole-of-government SBE framework.	Larger and more mature ocean economy, supported by strong maritime industries and greater policy continuity.	Canada's trajectory is more programmatic and emergent, whereas Norway's reflects deeper structural integration of ocean sectors.
Circular economy adoption	Strong and increasingly coherent policy architecture for plastics reduction, ghost gear management, and marine circularity.	Earlier and more operationalized circular measures, including waste-return, recycling, and port-based waste systems.	Canada is more visible in policy framing and international advocacy, whereas Norway shows greater implementation maturity.
Nature-based solutions	Strong progress in MPA expansion, habitat restoration, and Indigenous-led or co-managed conservation.	Established ecosystem-based marine management, but restoration and formal marine protection remain less developed.	Canada illustrates a restoration- and conservation-oriented pathway, whereas Norway reflects a more management-oriented approach.
Carbon alignment	Progress through shore power, early ferry electrification, and green shipping corridor participation, but implementation remains at an early stage.	Stronger maritime decarbonization through procurement, industrial policy, and rapid deployment of low-emission vessels.	Norway shows that carbon alignment is more effective when linked to coordinated industrial and deployment mechanisms.

Third, the two countries differ in how they prioritize sustainability within the blue economy. Norway's model emphasizes industrial modernization, maritime innovation, and low-emission transition within an already strong ocean economy. Canada's model places greater visible emphasis on restoration, protected areas, and inclusive governance, particularly through Indigenous participation in marine stewardship. These contrasting priorities are reinforced by different policy instruments: Norway relies more strongly on procurement, industrial incentives, and coordinated investment, whereas Canada relies more heavily on public programs, conservation mechanisms, and collaborative governance.

4.2. Structural challenges and unresolved tensions

Despite substantial progress, both countries face structural tensions that complicate their SBE trajectories. First, fossil-linked sectors remain influential in both political economies, creating potential inconsistencies between climate ambition and blue growth narratives. Second, circularity gaps persist, particularly in waste infrastructure, fisheries-related material flows, and end-of-life management for marine industries. Third, a persistent gap remains between policy adoption and operational implementation, particularly where private investment,

regulatory coordination, and supporting infrastructure are required [54–57]. These transitions may also distribute benefits, costs, and access to marine space unevenly among industries, coastal communities, Indigenous peoples, and small-scale users.

These tensions reflect broader critiques in the blue economy literature, which emphasize that the SBE is not only a technical sustainability framework but also a contested governance space in which priorities, sectors, and beneficiaries are continuously negotiated [1,2]. The comparison indicates that national SBE strategies can simultaneously advance transition objectives and reproduce established sectoral priorities. Industrial modernization may reduce emissions without necessarily resolving ecological protection concerns, while conservation progress does not automatically generate a coordinated low-carbon industrial transition. The central challenge is therefore to align economic, ecological, and social objectives rather than pursue them through parallel policy tracks. A further challenge concerns measurement. Differences in national definitions, reporting practices, temporal coverage, and data availability complicate direct cross-country comparison. This reinforces the need for caution when using KPI-based benchmarking in complex governance fields. As noted in the performance-measurement literature, indicator systems can structure comparison effectively, but they may also compress qualitative, relational, or context-dependent dimensions into simplified scores [25,26]. In the context of SBE, this is especially relevant because governance quality, justice, legitimacy, and long-term ecological outcomes are not always easily captured in comparable numerical terms.

Fig. 3 synthesizes these interconnected challenges, including policy coherence, circularity gaps, financing constraints, private-sector mobilization, and limitations in SBE data and metrics.

4.3. Mutual lessons and opportunities for collaboration

The comparison identifies opportunities for policy learning, although transfer should involve adaptation rather than direct replication. Canada could draw from Norway's use of public procurement, stable financing, and coordinated innovation policy to create predictable conditions for deploying low-emission vessels and port electrification at scale. Norway could draw from Canada's use of dedicated restoration funds, large-scale habitat programs, and collaborative conservation arrangements, while recognizing the legal and institutional specificity of Indigenous-led governance models.

The results also point to opportunities for bilateral and transnational collaboration. As active participants in international ocean governance, Canada and Norway could deepen cooperation on green shipping corridors, offshore renewables in cold climates, sustainable aquaculture innovation, and marine restoration knowledge exchange. Such cooperation should move beyond general knowledge exchange toward measurable pilot initiatives, joint monitoring arrangements, and shared research programs. In northern and Arctic contexts, these initiatives should include Indigenous, Sami, coastal, and industry representatives to strengthen both implementation capacity and social legitimacy [20,44,58].

4.4. International frameworks and implications for SBE benchmarking

Both countries broadly follow the direction of international frameworks that combine ocean-based development with conservation, circularity, climate alignment, and improved governance [5–7]. However, the comparison shows that these principles are implemented selectively rather than uniformly. National priorities, institutional capacity, and political-economic structures determine which dimensions receive stronger policy attention and which remain less developed. International SBE frameworks therefore provide a common direction, but not a standardized implementation pathway.

The findings also clarify the contribution and limitations of indicator-based SBE studies. Existing international assessments broaden evaluation beyond conventional economic output, but they can remain descriptive when indicators are not connected to the governance structures and policy instruments that shape performance. The present framework adds this interpretive dimension by linking KPI scores to policy sequencing, institutional configuration, and instrument choice. Nevertheless, the framework should not be interpreted as a universal SBE index. Justice, inclusion, institutional quality, scientific capacity, and distributional outcomes remain insufficiently represented, despite

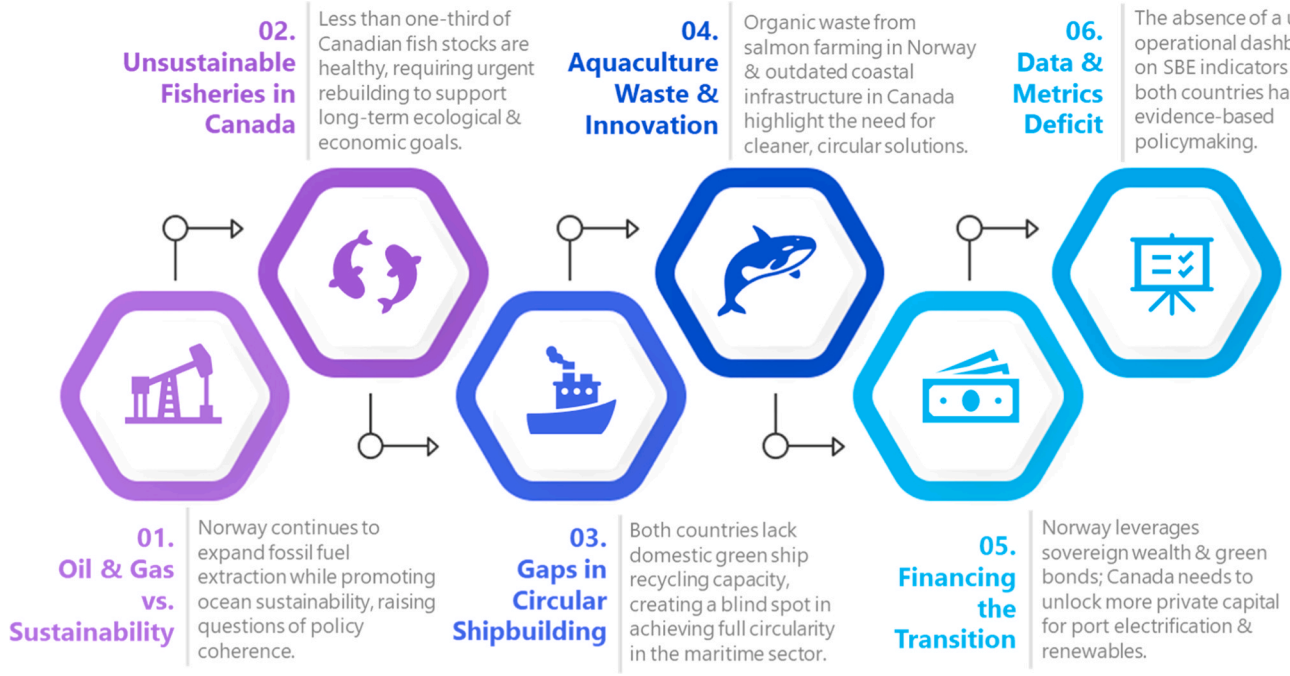


Fig. 3. Key system-level challenges facing Canada and Norway in the development of SBE. Source: Authors' synthesis based on the comparative findings and critical analysis presented in this study.

their growing importance in sustainable ocean governance [8]. Future applications should integrate these dimensions and test the framework across other Arctic and coastal states to distinguish broadly transferable indicators from those that remain dependent on national context.

5. Policy and research implications

The comparative findings have several implications for policy and research. To translate the comparative findings into actionable policy measures, Table 3 presents SMART-oriented recommendations specifying the proposed intervention, measurable indicators, responsible actors, rationale, and implementation timeframe.

The recommendations indicate that Canada's priority is to convert strategic ambition into coordinated implementation, particularly through green public procurement, port electrification, and harmonized marine-sector circularity measures. Norway's priority is to complement its maritime decarbonization leadership with stronger ecosystem restoration and effective marine protection. In both countries, implementation will require stable financing, regulatory coordination, measurable targets, and meaningful participation by coastal and Indigenous communities. From a research perspective, the comparison highlights the need for more harmonized SBE monitoring systems that integrate economic, environmental, carbon, social, and institutional indicators. Future studies should test the framework across a broader range of Arctic and coastal economies and assess whether the proposed indicators remain applicable under different governance, economic, and data conditions.

Fig. 4 translates the comparative findings into a policy-oriented roadmap linking governance reform, implementation instruments, stakeholder inclusion, monitoring, and long-term transition targets.

6. Conclusion

This study compared Canada's and Norway's SBE trajectories from 2015 to 2025 through a harmonized KPI-based framework covering four operational dimensions: blue economic growth, circularity, NBS, and carbon alignment. The results show that both countries are advancing sustainable ocean policy, but through different strengths and pathways. Norway performs more strongly in blue economic growth and maritime decarbonization, reflecting its mature ocean industries, coordinated governance, and technological leadership. Canada performs more strongly in NBS and shows strong momentum in circularity governance, particularly through marine conservation, restoration, and plastics policy.

These findings suggest that there is no single model of SBE implementation. Instead, SBE outcomes are shaped by differences in institutional legacy, economic structure, policy sequencing, and governance priorities. The comparison also shows that progress can be uneven across dimensions: leadership in industrial transition does not automatically translate into leadership in conservation and restoration, and vice versa. In that sense, the study supports the broader literature that treats the blue economy as a contested and multi-dimensional governance space rather than a uniform policy template.

Methodologically, the paper contributes an operational benchmarking framework that can support structured comparison across national contexts. At the same time, it also highlights the limits of KPI-based assessment in complex policy fields marked by data heterogeneity, interpretive scoring, and under-measured social dimensions. The four dimensions used here provide a workable comparative basis, but they do not capture the full normative breadth of the SBE, particularly regarding justice, inclusion, and institutional quality. Future research should therefore extend comparative SBE benchmarking by integrating social, institutional, and scientific indicators more explicitly.

Canada and Norway illustrate two influential but distinct pathways toward sustainable ocean governance. The findings support accelerated green-vessel procurement, port electrification, and marine-sector circularity in Canada, alongside expanded ecosystem restoration and effective marine protection in Norway. Both countries would also benefit from stronger bilateral cooperation on green shipping, Arctic innovation, and harmonized SBE monitoring. However, the study is limited by its two-country design, heterogeneous national datasets, and partly interpretive KPI scoring. The framework should therefore be tested and adapted in countries with different institutional, economic, and data capacities before broader generalization.

CRediT authorship contribution statement

Amine el Mahdi Safhi: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nezha Mejjad:** Writing – review & editing, Validation. **Mounir Ghribi:** Validation, Supervision. **Shima Pilehvar:** Validation, Supervision, Resources, Project administration, Funding acquisition.

Table 3
SMART-oriented policy recommendations for advancing SBE implementation in both countries.

Country	Policy recommendation	Proposed KPI or target	Lead actors	Rationale	Timeframe
Canada	Establish a coordinated zero-emission vessel procurement roadmap for ferries and public marine fleets.	Share of new publicly procured vessels that are zero-emission or zero-emission-ready; annual fleet-emission reductions.	Federal and provincial governments, ferry operators, port authorities.	Converts climate commitments into predictable demand for clean maritime technologies.	Roadmap by 2027; progressive implementation to 2030.
Canada	Accelerate shore-power and low-carbon port infrastructure deployment.	Proportion of major ports equipped with shore power; annual vessel calls using shore-side electricity.	Transport Canada, port authorities, utilities, infrastructure agencies.	Reduces emissions at berth and supports green shipping corridors.	Priority ports by 2028; broader deployment by 2030.
Canada	Introduce coordinated extended producer responsibility for fishing gear and other marine-sector materials.	Recovery and recycling rates; quantity of ghost gear retrieved; number of provinces adopting harmonized requirements.	Federal, provincial, and territorial governments; fisheries organizations; producers.	Addresses persistent circularity gaps and fragmented waste governance.	Framework by 2027; national implementation by 2030.
Norway	Expand restoration of kelp forests, wetlands, salmon habitats, and other blue ecosystems.	Annual area restored; ecological condition indicators; carbon and biodiversity benefits documented.	National environmental agencies, municipalities, research institutes, coastal stakeholders.	Complements Norway's ecosystem-based management with measurable restoration outcomes.	National targets by 2027; scaled implementation to 2030.
Norway	Strengthen effective marine protection and management in biodiversity-sensitive areas.	MPA coverage, proportion with operational management plans, and indicators of ecological effectiveness.	National ministries, environmental agencies, fisheries authorities, Sami and coastal representatives.	Improves the ecological credibility of Norway's SBE pathway and supports the 2030 conservation agenda.	Progressive implementation through 2030.
Both	Develop a harmonized SBE monitoring framework and bilateral green-shipping initiatives.	Common annual dashboard for economic, circularity, NBS, carbon, and social indicators; establishment of at least one bilateral pilot corridor.	National governments, statistical agencies, ports, industry, Indigenous and coastal organizations.	Improves comparability, policy learning, and cooperation on Arctic and low-carbon maritime transitions.	Dashboard by 2027; pilot initiative by 2028–2030.

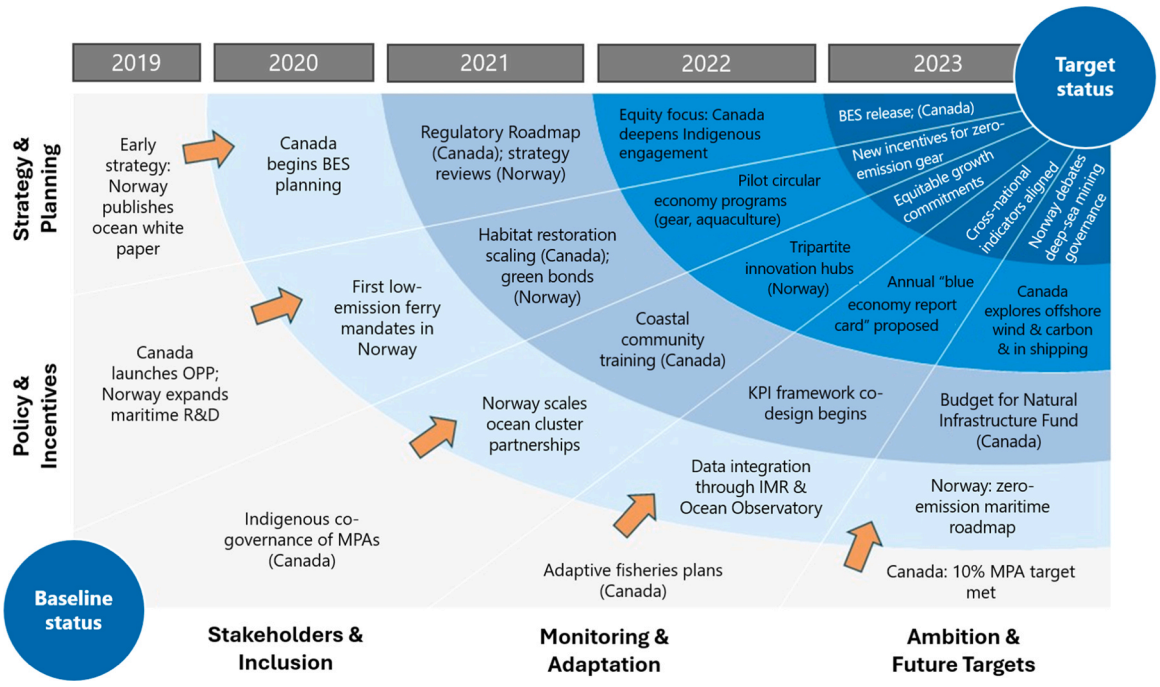


Fig. 4. Policy roadmap for advancing SBE transitions in Canada, Norway, and comparable Arctic Ocean economies. Source: Authors' synthesis based on the comparative findings and policy implications developed in this study.

Mahdi Kioumarsi: Validation, Supervision, Resources, Project administration, Funding acquisition.

Conflict of Interest

The authors declare no conflict of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.marpol.2026.107298.

Data Availability

No data was used for the research described in the article.

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